Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We use a fixed one-minute calendar-time grid from 09:30 to 15:59 for each trading day. This corresponds to 390 one-minute observations per normal trading day. Since a stock may not trade in every minute, we handle missing prices using previous-tick interpolation within each trading day only. This means that if AAPL has no observed price in a given minute, we carry forward the most recently observed price from the same trading day. We do not carry prices across trading days, and we do not backfill prices before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, we sample prices on a fixed grid relative to the market open. For example, with 10-minute sampling, prices are taken at 09:30, 09:40, 09:50, and so on. This differs from a bucket-based approach, where the last observed price within each Δ-minute interval is used. We prefer the fixed-grid approach because it makes the sampling rule explicit and because it will later allow us to align intraday returns across all 30 stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where (r_{d,j}^{( )}) is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

The volatility signature plot shows that the average realized variance is highest at the shortest sampling intervals and falls substantially when moving from 1-minute to 5-minute sampling. This is

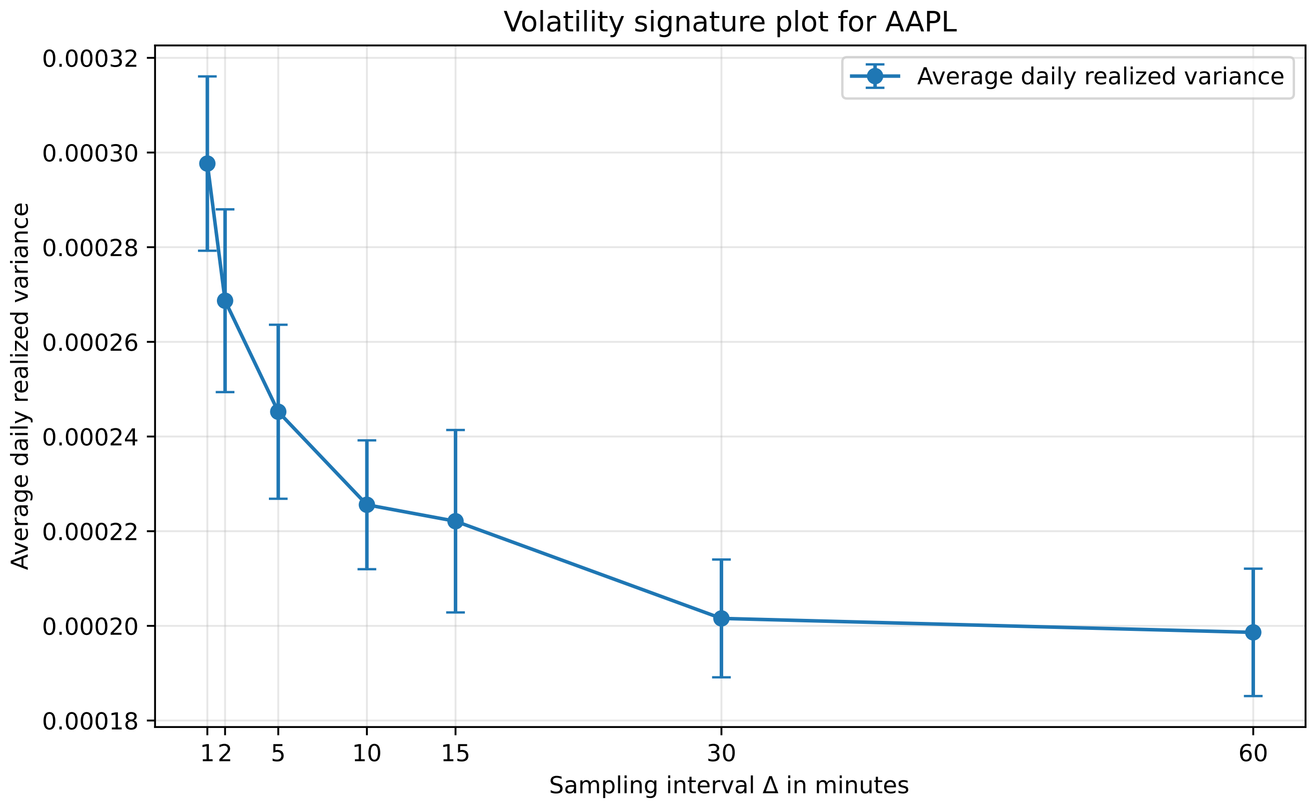

Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

consistent with the presence of market microstructure noise at very high frequencies. In the lecture slides, the observed log price is written as the latent efficient log price plus a noise component, $Y_{i\Delta_n} = X_{i\Delta_n} + U_{i\Delta_n}$, and the noise-driven fraction of return variance increases as Δ_n becomes small. Sampling less frequently therefore reduces the influence of bid-ask bounce, price discreteness and stale prices.

We choose the standard realized variance estimator based on 5-minute log returns for the remainder of the analysis:

$$RV_d^{(5)} = \sum_j \left(r_{d,j}^{(5)} \right)^2 .$$

This choice is also consistent with the empirical realized-volatility literature, where 5-minute returns are commonly used as a practical compromise between using high-frequency information and limiting market microstructure noise. Andersen and Bollerslev discuss the use of high-frequency intraday returns to obtain improved volatility measurements and refer to empirical analysis based on five-minute returns. At the same time, 5-minute sampling leaves approximately $390/5 - 1 = 77$ intraday returns per trading day. Since the subsequent realized covariance matrix is 30-dimensional, this helps avoid the rank problem emphasized in the lecture slides: when the return matrix has $T < N$, the sample covariance matrix is singular and cannot be inverted. Thus, 5-minute sampling is preferred to lower-frequency choices such as 15, 30 or 60 minutes, which would leave fewer intraday returns for estimating the 30×30 covariance matrix.

Exercise 2a: Daily realized covariance matrices

We estimate the daily realized covariance matrix for the 30 DJIA stocks using the standard realized covariance estimator

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j},$$

where $r_{d,j}$ denotes the 30×1 vector of synchronized intraday log returns at interval j on trading day d. This estimator is the multivariate analogue of the realized variance estimator used in Exercise 1.

Following Exercise 1, covariance is estimated using synchronized 5-minute returns. The same sampling frequency is used throughout the remainder of the analysis, ensuring consistency with the realized variance estimator selected based on the volatility signature plot.

To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day and missing observations are handled using previous-tick interpolation (forward filling). Returns are then computed on a common 5-minute grid, ensuring that the covariance matrix is based on synchronized return vectors.

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i, the diagonal element is

$$RC_{d,ii} = \sum_{j=1}^{M_d} r^2_{d,j,i}.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

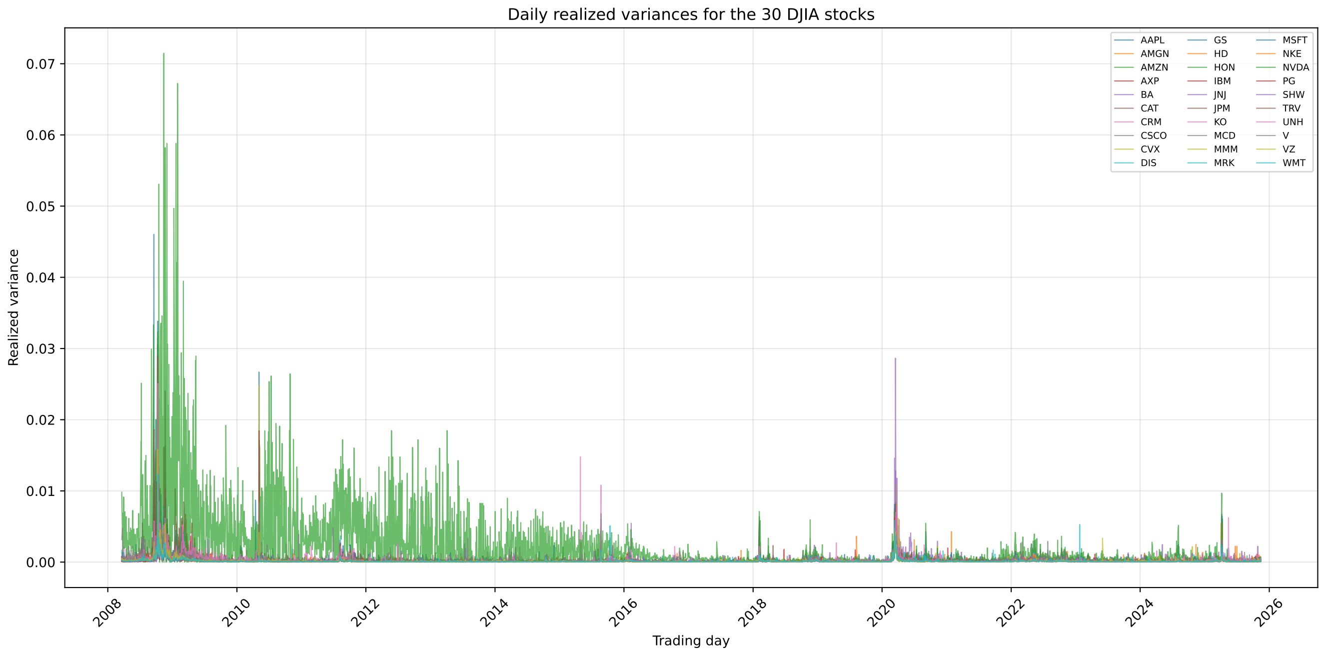

Figure 2: Daily realized variances for the 30 DJIA stocks.

The realized variance series exhibit clear volatility clustering, with periods of high volatility followed by further periods of high volatility and vice versa. Several stocks experience large volatility spikes during periods of market stress, indicating that volatility tends to increase simultaneously across assets.

The figure also shows substantial time variation in realized variance, suggesting that volatility is far from constant over time. While the overall patterns are similar across stocks, some stocks display persistently higher realized variance than others, reflecting differences in underlying risk characteristics.

By sampling at a one-minute frequency instead, the off-diagonal elements of RC_d would likely be more affected by market microstructure noise. As discussed in the lecture slides, Andersen and Bollerslev (1998), and Barndorff-Nielsen et al. (2011), covariance estimates at very high frequencies can suffer from the Epps effect, whereby the measured dependence between assets is biased downward. Although the DJIA data are recorded on a common one-minute grid and therefore largely mitigate the non-synchronous trading problem, one-minute returns remain more sensitive to bid-ask bounce and price discreteness. Consequently, the off-diagonal elements of RC_d may become noisier and potentially lower than those obtained from synchronized five-minute returns. This provides additional support for the five-minute sampling frequency selected in Exercise 1.

Exercise 3: Covariance estimators

For each day t in the second half of the sample, we construct two estimators of today's covariance matrix using only information available up to day $t-1$.

Exercise 3a: High-frequency random walk estimator

The high-frequency estimator is defined as

$$\hat{\Sigma}_t^{HF} = RC_{t-1}.$$

Thus, yesterday's realized covariance matrix is used as today's covariance forecast. This implies that the covariance matrix follows a random-walk process, such that changes in covariance are unpredictable and arise only from new shocks or information. Consequently, yesterday's realized covariance matrix provides the best forecast of today's covariance matrix. This assumption is motivated by the persistence and clustering typically observed in volatility and covariance dynamics.

Exercise 3b: Ledoit-Wolf shrinkage estimator

For the Ledoit-Wolf estimator, we first construct daily close-to-close log returns for all 30 stocks using the last observed intraday price on each trading day. Meaning, the estimator relies on daily return information, where the high-frequency estimator is constructed from realized covariance matrices based on intraday returns.

For each evaluation day t, the Ledoit-Wolf covariance matrix is estimated from a rolling window of $W = 252$ daily returns ending on day $t - 1$. Let S_t denote the sample covariance matrix computed from these W daily returns. The Ledoit-Wolf shrinkage estimator is then given by

$$\hat{\Sigma}_t^{LW} = \delta_t F_t + (1 - \delta_t) S_t,$$

where F_t is a structured shrinkage target and $\delta_t \in [0, 1]$ denotes the shrinkage intensity. Thus, the estimator shrinks the sample covariance matrix towards a more stable target in order to reduce estimation error. We set $W = 252$, corresponding to approximately one trading year. This provides a balance between estimation precision and adaptability: a shorter window would produce noisier covariance estimates that are more sensitive to individual observations, whereas a much longer window would react more slowly to changes in the covariance structure by placing greater weight on older data. A one-year rolling window is therefore a commonly used compromise in empirical finance when estimating covariance matrices from daily returns.Only information available before day t is used. The resulting covariance matrix is stored as $\hat{\Sigma}_t^{LW}$, and its diagonal elements are saved for the comparison in Exercise 3c.

Exercise 3c: Comparing realized and Ledoit-Wolf variances

To compare the smoothing properties of the two estimators, we focus on the diagonal elements of the covariance matrices. The diagonal elements of $\hat{\Sigma}_t^{LW}$ are the Ledoit-Wolf implied variance forecasts, while the diagonal elements of RC_t are the realized variances computed from intraday returns in Exercise 2b.

Thus, for each stock i, we compare

$$\hat{\Sigma}^{LW}_{t,ii} \quad \text{and} \quad RC_{t,ii}.$$

The Ledoit-Wolf estimator is based on a rolling window of daily close-to-close returns and shrinkage, while RC_t is based on high-frequency intraday returns. The comparison therefore illustrates how much smoothing is introduced by the Ledoit-Wolf estimator relative to the more volatile realized variance series.

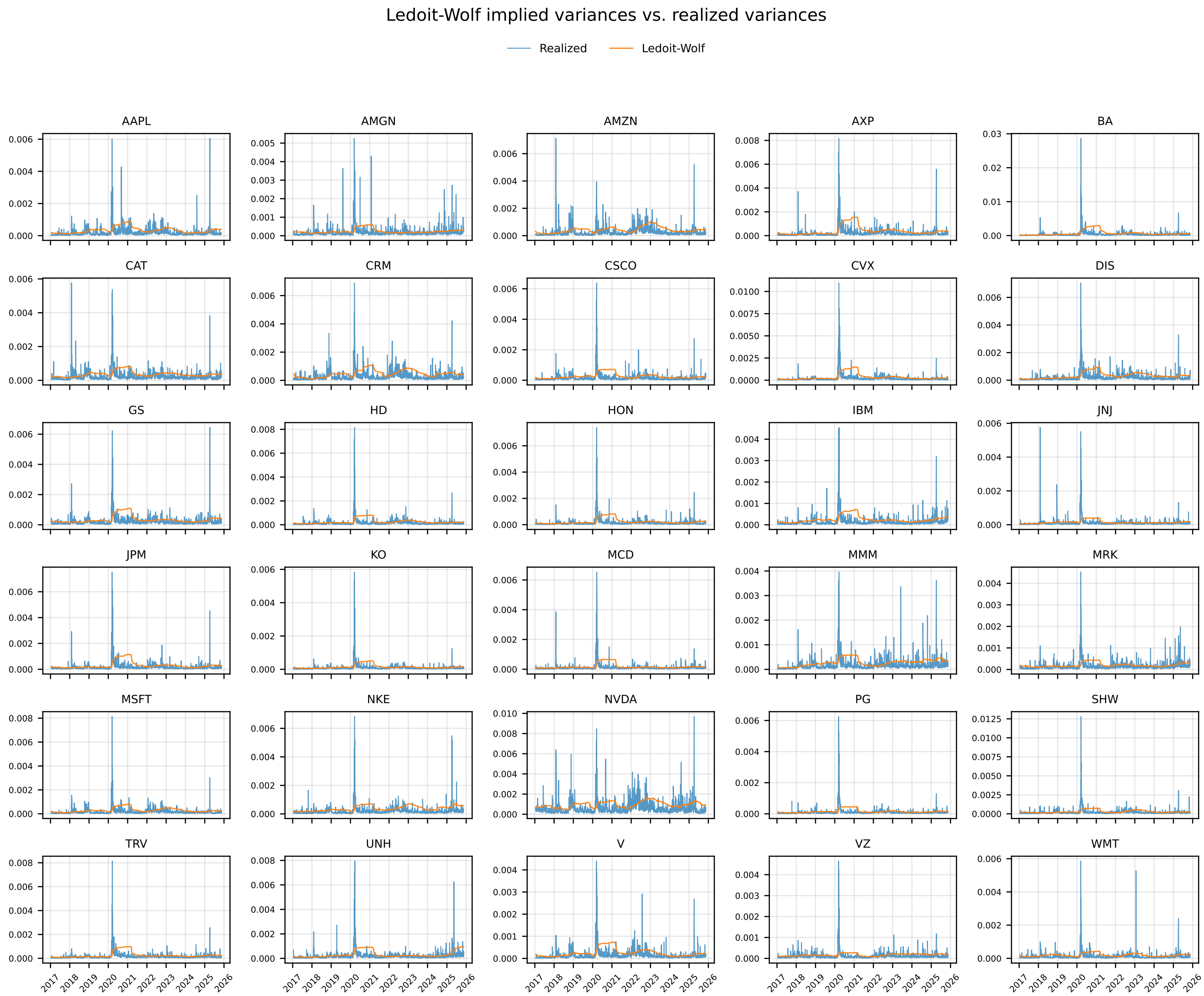

Figure 3: Ledoit-Wolf implied variances and realized variances for the 30 DJIA stocks. The Ledoit-Wolf estimates are based on a 252-day rolling window of daily close-to-close returns, while realized variances are the diagonal elements of the daily realized covariance matrices.

The figure shows that the realized variances are substantially more volatile than the Ledoit-Wolf implied variances. Because they are computed directly from intraday returns, the realized variances contain sharp daily spikes and respond immediately to short-run volatility shocks. A pronounced increase in variance is observed across almost all stocks around early 2020, reflecting the heightened market uncertainty during the COVID-19 pandemic. Both estimators capture this increase in risk, but the Ledoit-Wolf estimates remain considerably smoother because they are based on a 252-day rolling window of daily close-to-close returns and apply shrinkage to reduce estimation error. As argued by Ledoit and Wolf (2003, 2004), shrinkage pulls noisy and extreme covariance estimates towards a more stable target. Consequently, the realized variance estimator reacts immediately to new information, whereas the Ledoit-Wolf estimator provides a more stable estimate of risk but responds more gradually to sudden changes in volatility.

Exercise 4: Minimum-variance portfolios

Using the covariance matrix forecasts constructed in Exercise 3, we form daily minimum-variance portfolios for the evaluation period. Specifically, we compute portfolio weights based on both the high-frequency random-walk estimator, $\hat{\Sigma}_t^{HF}$, and the Ledoit-Wolf shrinkage estimator, $\hat{\Sigma}_t^{LW}$.

For a given covariance matrix estimate $\hat{\Sigma}_t$, the minimum-variance portfolio is obtained by solving

$$\omega_{mvp,t} = \arg\min_{\omega_t} \ \omega_t' \hat{\Sigma}_t \omega_t \quad \text{s.t.} \quad \iota' \omega_t = 1.$$

Solving the first-order conditions yields the closed-form solution

$$\omega_{mvp,t} = \frac{\hat{\Sigma}_t^{-1} \iota}{\iota' \hat{\Sigma}_t^{-1} \iota},$$

where ι denotes a vector of ones.

The resulting portfolio weights reflect how the different covariance estimators translate into portfolio allocations. Since the high-frequency estimator is based on the most recent realized covariance matrix, it is expected to react more strongly to short-run fluctuations in covariance. In contrast, the Ledoit-Wolf estimator uses a rolling window of daily returns and shrinkage, which may lead to more stable covariance estimates and therefore more stable portfolio weights. We compare the resulting portfolio allocations, identify the most extreme positions, and evaluate the average turnover between rebalancing dates.

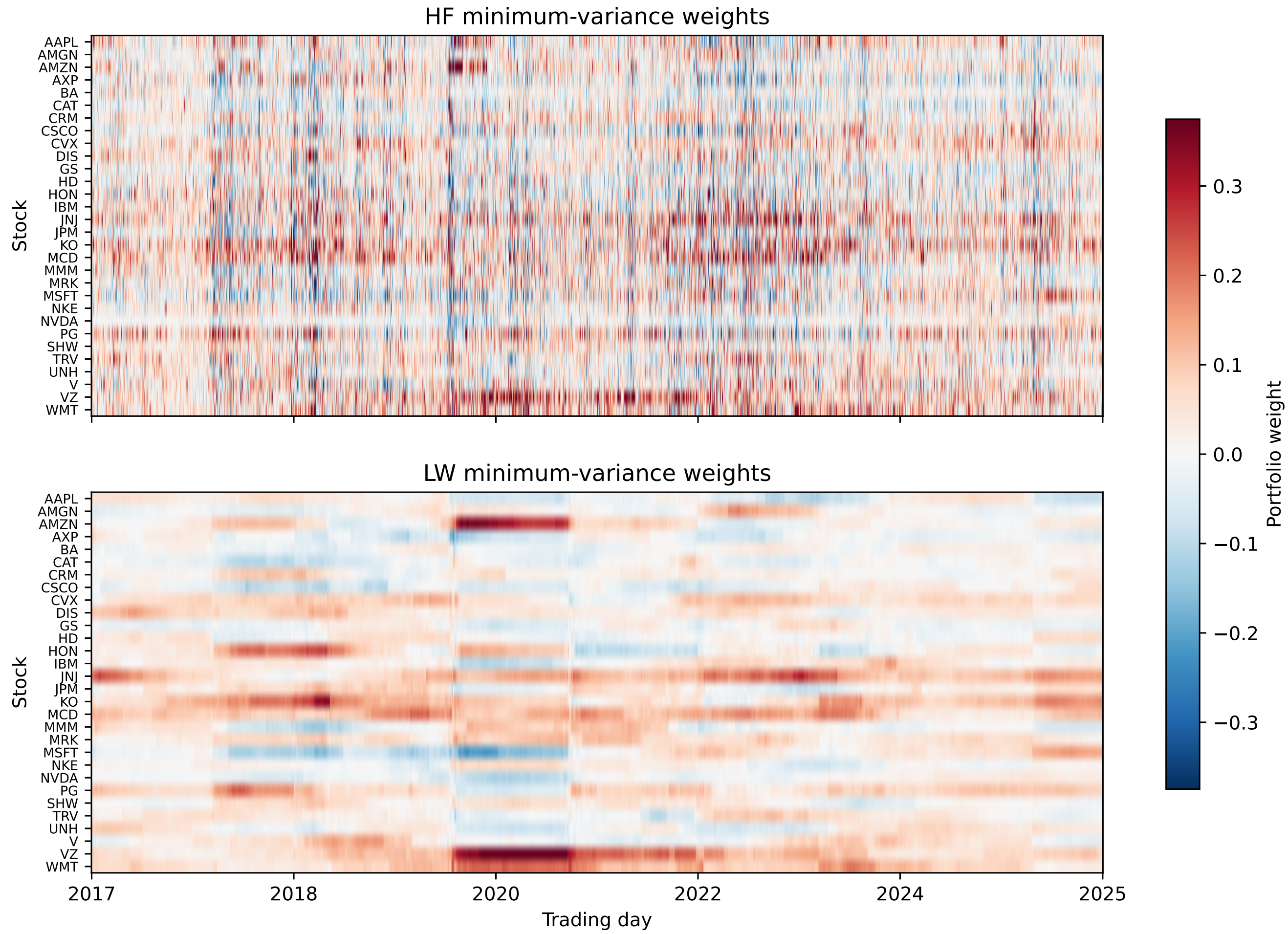

Figure 4: Minimum-variance portfolio weights for the high-frequency and Ledoit-Wolf covariance estimators. Rows are stocks, columns are rebalancing dates. Red indicates long positions, blue indicates short positions, and white indicates weights close to zero. Both panels share the same colour scale, capped at the 99th percentile of absolute weights so that typical variation is visible.

	Trading day	Strategy	Stock	Weight (%)	Absolute weight
0	2021-10-05	HF	VZ	148.10	1.4810
1	2020-03-30	HF	AMZN	133.97	1.3397
2	2020-03-09	HF	AMZN	119.76	1.1976
3	2020-03-19	HF	AAPL	119.02	1.1902
4	2021-10-01	HF	VZ	118.94	1.1894
5	2020-03-04	HF	HD	-116.91	1.1691
6	2020-03-27	HF	AMZN	110.87	1.1087
7	2020-04-13	HF	AMZN	110.12	1.1012
8	2023-01-05	HF	JNJ	100.63	1.0063
9	2021-10-07	HF	VZ	100.23	1.0023

	Strategy	Average turnover (%)	Median turnover (%)	Max turnover (%)	Rebalances
0	HF	349.65	328.44	934.56	2222
1	LW	7.12	4.99	98.45	2222

weights_long[“weight”].describe() ““

References